

Python, NLP with spaCy, and LLMs using LangChain

Course Outline

Module 1: Python Programming Essentials

Python Basics

- Variables, Data Types, Type Casting
- Operators and Expressions
- Input/Output and String Formatting
- Conditional Statements (if, elif, else)

Data Structures

- Lists, Tuples, Sets, Dictionaries
- List comprehensions
- Iteration and Looping (for, while, break, continue)

Functions and Modules

- Defining and calling functions
- *args and **kwargs
- Importing modules and pip package management
- Creating reusable scripts

Working with Files

- Reading and writing files
- Context managers (with statement)
- Working with CSV and JSON

Object-Oriented Programming (OOP)

- Classes and Objects
- Inheritance and Polymorphism
- Encapsulation and abstraction

Module 2: Introduction to NLP using spaCy

NLP Foundations

- What is NLP?
- Key tasks: Tokenization, POS Tagging, NER, Dependency Parsing
- NLP use cases

Getting Started with spaCy

- Installing spaCy and loading language models
- Tokenization and Linguistic Features
- Part-of-Speech (POS) Tagging
- Named Entity Recognition (NER)

spaCy Text Processing

- Sentence Segmentation
- Dependency Parsing and Tree Visualization
- Rule-based Matching with Matcher and PhraseMatcher

Text Preprocessing with spaCy

- Stopword removal
- Lemmatization
- Custom pipelines and adding components

Practical NLP Applications

- Resume parser
- Legal/medical entity extractor
- Document classification pipeline

Module 3: Working with LLMs using LangChain

Introduction to LLMs

- What is a Large Language Model?
- GPT, Claude, LLaMA, Mistral overview
- Use cases: Chatbots, summarization, QA systems

LangChain Fundamentals

- What is LangChain and why use it?
- Core concepts: PromptTemplates, Chains, Agents, Tools
- Setting up LangChain and API keys (OpenAI, HuggingFace)

Document Handling & Embeddings

- Document loaders (PDF, CSV, web)
- Chunking strategies
- Generating and storing embeddings
- Vector stores: FAISS, Chroma, Pinecone

Retrieval-Augmented Generation (RAG)

- What is RAG?
- Implementing a RAG pipeline with LangChain
- Combining vector search with LLM response

Building LLM Applications

- Chatbot with memory
- AI Q&A system for custom PDFs
- LangChain Agents using tools (Calculator, Search, etc.)
- Streaming outputs with CallbackHandlers

Tools & Interfaces

- Streamlit or Gradio UI for LLM apps
- LangServe for LangChain APIs
- Deploying on cloud (optional)

Capstone Project Ideas

- NLP Resume Analyzer using spaCy + Streamlit
- Chat with Your Docs using LangChain + FAISS + OpenAI
- LLM-Powered Language Tutor using GPT + LangChain